Seismology

Pupil sheet

Name: _____ **Date:** _____

Safety: I strike the table or the floor, never the box; only one person pushes the table, gently.

1. What I see on the screen

The screen shows five cards and an **event counter**. The **Shaking** card (in g) shows $|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$ — **and it includes gravity**, so at rest it reads about 1 g, not zero. **It is not an earthquake's magnitude.**

The counter, though, subtracts gravity first and counts a shake when the departure from rest passes 0.2 g (the dashed red line on the chart). A shake is counted **once**, however long it lasts.

The counter is not exported: I write it on paper, before and after each interval.

2. What I do on my own

First I predict. Five vertical palms will raise the counter by about _____; five sideways pushes, just as strong, by about _____. One long two-second shake, by about _____.

because _____

Test 1 — direction. Five knocks of the same strength in each direction, with silence between the sets.

Direction	Before	After	Difference	Seconds
Vertical				
Horizontal				

Test 2 — duration. Same direction, same strength. Once one long shake, once five short knocks.

Kind	Before	After	Difference	Seconds
One shake, 2 s				
Five short knocks				

The counter counted: ☐ the seconds of shaking ☐ the shakes, one each

Test 3 — two intervals. Two different places, or two different times of day. **The same duration for both**, or the comparison means nothing.

Place / time	Before	After	Difference	Minutes

3. The conclusion I write

I predicted that, and the data show that _____

_____

Direction did not matter because _____

_____

The counter does not count earthquakes, but _____

_____

Exit ticket

The counter reached 34. What did it count — and what did it not?

_____